

Ground then Navigate: Language-guided Navigation in Dynamic Scenes

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CERTIFICATE

It is certified that the work contained in this thesis, titled “Ground then Navigate: Language-guided Navigation in Dynamic Scenes” by Varun Chhangani, has been carried out under my supervision and is not submitted elsewhere for a degree.

August 1, 2026

Adviser: Dr. Vineet Gandhi

उद्यमेन हि सिध्यन्ति कार्याणि न मनोरथैः।
न हि सुप्तस्य सिंहस्य प्रविशन्ति मुखे मृगाः॥

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Abstract

We investigate the Vision-and-Language Navigation (VLN) problem in the context of autonomous driving in outdoor settings. Existing VLN methods typically pose navigation as a node-selection problem on a discrete connected graph of the environment, or as grounding on a single static image. Both assumptions break down in dynamically changing outdoor scenes: a discrete map is often unavailable, and the region referred to by a language command may not yet be visible, may be occluded, or may leave the frame as the vehicle moves.

This thesis addresses these limitations by explicitly grounding the navigable regions corresponding to a textual command. At each timestamp, the model predicts a segmentation mask for the intermediate or final navigable region. The approach moves toward continuity in action space, provides interpretability through visual feedback, and supports finer manoeuvres such as “park between the two cars”.

To enable training and live evaluation, we introduce CARLA-NAV, a novel meta-dataset built on the CARLA simulator. It comprises pre-recorded training sequences and a live environment for validation and testing, spanning multiple maps, weather conditions, and dynamic traffic. We further propose a multi-task network that jointly predicts navigable-region and short-term future-trajectory masks from multi-frame visual features, a historical contextual trajectory, and the language command, using a CLIP backbone.

Extensive qualitative and quantitative experiments demonstrate that combining multiple frames with historical trajectory context substantially improves Task Completion, Fréchet Distance, and normalized Dynamic Time Warping over single-image and non-contextual baselines. To the best of our knowledge, this is among the first attempts at live language-guided navigation in dynamic outdoor environments.

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List of Acronyms

BERT	Bidirectional Encoder Representations from Transformers
CARLA	Car Learning to Act
CIRL	Controllable Imitative Reinforcement Learning
CLIP	Contrastive Language–Image Pretraining
CNN	Convolutional Neural Network
DUET	Dual-scale Graph Transformer
EnvDrop	Environmental Dropout
GPS	Global Positioning System
IMU	Inertial Measurement Unit
LiDAR	Light Detection and Ranging
LSTM	Long Short-Term Memory
nDTW	Normalized Dynamic Time Warping
NLP	Natural Language Processing
R2R	Room-to-Room
REC	Referring Expression Comprehension
RIS	Referring Image Segmentation
RNR	Referring Navigable Regions
RNN	Recurrent Neural Network
SLAM	Simultaneous Localization and Mapping
TC	Task Completion
ViT	Vision Transformer
VLN	Vision-and-Language Navigation

Chapter 1

Introduction

1.1 Scope of the Thesis

The focus of this thesis is Vision-and-Language Navigation (VLN) for autonomous driving in outdoor settings. Humans routinely navigate from linguistic descriptions of the environment—associating visual elements with words such as “the black car on the left” or “the bus stand after the traffic light.” Transferring this capability to an autonomous vehicle requires grounding language in a continuously changing visual scene and converting that grounding into actionable control.

Existing VLN methods fall short in this setting. Indoor approaches often assume a known map that can be discretized into a navigation graph [3, 4]. Outdoor methods frequently restrict the agent to a small discrete action set on pre-recorded street views [5, 6]. Single-image visual grounding methods for navigable regions [1] assume that the referred region is always visible in the current frame. None of these assumptions hold for live outdoor driving, where the environment is unknown, dynamic, and where commands routinely refer to regions that are not yet in view.

This thesis reformulates the Referring Navigable Regions (RNR) task for dynamic outdoor environments: at each timestamp the model predicts a segmentation mask corresponding to an intermediate or final navigable region, which is then passed to a local planner for continuous control. We also introduce CARLA-NAV, a meta-dataset for training and live evaluation in the CARLA simulator [7].

1.2 Research Problems

A *Grounding navigable regions when the referred goal is not visible.*

Single-image grounding methods [1, 8, 9] fail when the region described by the command is occluded, out of frame, or not yet visible. Everyday commands such as “take a right

once you see the traffic signal” require predicting intermediate navigable regions until the final goal comes into view.

B *Moving beyond discrete action spaces for outdoor VLN.*

Discretizing actions (FORWARD, LEFT, RIGHT, STOP) or environment graphs limits fine manoeuvres such as “park between the two cars on the right” [5, 10]. Outdoor VLN needs continuous control over any drivable region on the road.

C *Exploiting temporal and trajectory context for live navigation.*

Static-image methods lack short-term and long-term temporal context. For multi-step commands (e.g., “turn left and park near the blue dustbin”), the model must know which part of the instruction has already been executed. Historical trajectory context and multi-frame visual features are essential for live inference in a changing scene.

1.3 Motivation

1.3.1 Limitations of Single-Image Visual Grounding

Referring Expression Comprehension (REC) [9, 11] and Referring Image Segmentation (RIS) [8] localize objects from language, but object boxes or masks do not answer “which region on the road to drive to.” Referring Navigable Regions (RNR) [1] addresses this by predicting a navigable region on the road for a linguistic command. However, RNR is limited to static images from pre-recorded sequences and assumes the target is always visible. Figure 1.1 illustrates a characteristic failure: the model correctly grounds a visible black car, yet completely confuses a left-turn command that does not exist in the current view.

1.3.2 Limitations of Graph-Based and Discrete-Action VLN

Indoor VLN often assumes a known topology and models navigation as graph traversal [3, 12, 13]. Outdoor datasets such as Touchdown [5] operate on Google Street View with four discrete actions. These designs preclude fine-grained outdoor manoeuvres and make live evaluation in a fully navigable dynamic environment difficult. Prior CARLA-based language work has also been limited in action diversity [14], and Talk2Car focuses on localizing referred objects rather than navigating to regions [15].

1.3.3 Need for a Live Outdoor Benchmark

Evaluating on frames from pre-recorded videos does not measure whether an agent can complete an entire episode from start to finish in a changing scene. A dataset that supports



Figure 1.1: A major limitation of single image-based grounding methods is that they fail if the language command is not immediately visible, which restricts these methods for VLN. Here we show an example using the RNR model [1] on an image from their dataset. The model accurately localizes the black car (middle), but completely confuses when asked to predict a left turn that does not exist in the current view.

both supervised training on pre-recorded sequences and live validation in simulation is required to study language-guided outdoor navigation end to end.

1.4 Contributions

This thesis makes the following contributions:

1. We present CARLA-NAV, a vision-language navigation toolkit and meta-dataset on the CARLA simulator that enables fine-grained, language-based navigational manoeuvres with continuous control via a local planner.
2. We propose a multi-task network for per-frame navigable-region prediction and short-term future trajectory prediction in dynamic outdoor environments. Both outputs are interpretable segmentation masks.
3. We perform real-time (live) navigation in CARLA with a diverse set of linguistic commands, using predicted masks as targets for the local planner and an area-based stopping criterion.
4. We provide extensive qualitative and quantitative ablations showing that multiple frames and historical contextual trajectory are both necessary for effective outdoor VLN.

1.5 Thesis Layout

- C1 This chapter introduces the scope of the thesis, the research problems, motivation from prior work, and the main contributions.
- C2 Chapter 2 reviews background on autonomous navigation, computer vision, natural language processing, visual–language embeddings, language-guided navigation datasets and methods, and the CARLA simulator.
- C3 Chapter 3 describes the CARLA-NAV data collection process, dataset characteristics, and challenges encountered during curation.
- C4 Chapter 4 presents the problem formulation, multi-task network, live-navigation pipeline, evaluation metrics, and experimental results.
- C5 Chapter 5 concludes the thesis and outlines directions for future work.

Chapter 2

Background

In this chapter, we review the foundational concepts that underpin the rest of the thesis. We begin with autonomous navigation as it relates to self-driving vehicles, then cover the computer vision and natural language processing building blocks needed for language-guided driving. We discuss visual–language embeddings, focusing on Contrastive Language–Image Pretraining (CLIP), and survey Vision-and-Language Navigation (VLN) datasets and methods. We close with the CARLA simulator and a summary that connects this background to Chapters 3 and 4.

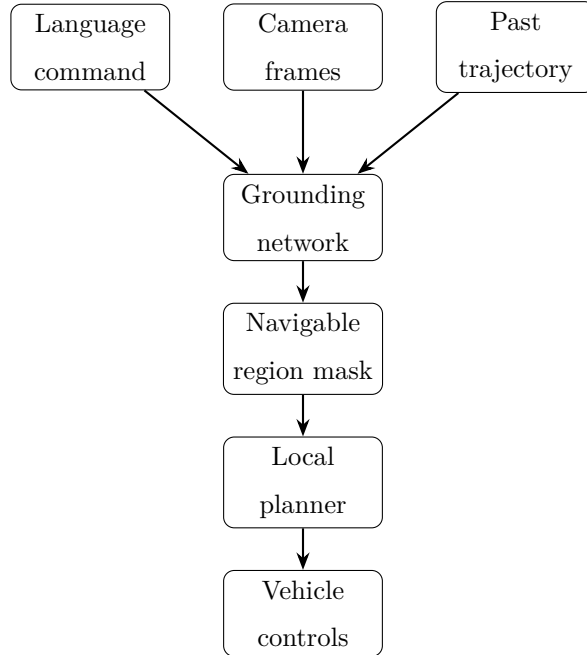


Figure 2.1: High-level Vision-and-Language Navigation (VLN) setup considered in this thesis: a language command and visual observations are grounded into a navigable region, which a local planner executes in continuous action space.

2.1 Introduction to Autonomous Navigation

Autonomous navigation is the science of enabling a robot or a vehicle to move through its environment without continuous human control [16]. It is fundamental to self-driving cars, delivery robots, and related systems [17, 18]. The classical pipeline comprises several interconnected stages: perception, localization, mapping, path planning, and control [19].

For road vehicles, the importance of autonomous navigation can be summarised as follows:

1. **Safety and efficiency:** Reducing human error while improving traffic flow and mobility [20].
2. **Real-time operation:** Perception and planning must run at rates compatible with driving dynamics [21].
3. **Coexistence with humans:** Agents must interact safely with human-driven vehicles, pedestrians, and cyclists in mixed traffic [20].
4. **Open research problems:** Dynamic scenes, language-conditioned goals, and continuous control remain active research areas that motivate the work in this thesis.

Perception uses sensors such as cameras, LiDAR, radar, and ultrasonics to observe the environment [22, 17]. **Localization** estimates the vehicle’s pose within that environment [16], using odometry, inertial measurement units (IMUs), GPS, or Simultaneous Localization and Mapping (SLAM) [23]. **Mapping** builds a representation of the surroundings—from a simple 2D occupancy grid to a rich 3D map—for planning and obstacle avoidance [24, 25]. **Path planning** computes a route from the current pose to a goal while accounting for obstacles [26, 27]. **Control** executes the plan by issuing actuator commands and correcting deviations from feedback [28].

Common navigation paradigms include:

- **Reactive navigation:** real-time responses without a global map [29, 30]; fast in dynamic scenes, but not always globally efficient.
- **Map-based navigation:** planning on a known map [16, 25]; efficient when the map is accurate, brittle when the world changes.
- **SLAM:** jointly estimating pose and map in an unknown environment [23, 31, 32], with LiDAR-based and visual variants [33].

In this thesis, the local planner plays the role of path planning and control, while our contribution focuses on perception and language understanding: predicting which road region the vehicle should drive toward given a linguistic command (Figure 2.1).

2.2 Basics of Computer Vision

Computer vision develops methods for acquiring, processing, and interpreting digital images so that machines can extract information useful for decisions [34]. Image data may come from single cameras, multi-camera rigs, RGB-D sensors, or LiDAR. Sub-domains include object detection, semantic segmentation, tracking, reconstruction, and related tasks. In autonomous driving, vision is used to detect lanes, vehicles, pedestrians, and traffic signals, and to understand the road scene [35].

2.2.1 Basic Methods in Computer Vision

Methods especially relevant to language-guided driving include:

- **Object Detection:** localising specific objects such as vehicles, pedestrians, and signs [36, 37].
- **Image Segmentation:** partitioning an image into regions that are easier to analyse [38], e.g. road versus sidewalk [39].
- **Semantic Segmentation:** assigning a class label to every pixel, which is useful for identifying drivable surface, pedestrians, and vehicles [40].

Our approach treats navigable-region prediction as a dense segmentation problem: the network outputs a pixel-level mask of the region to drive toward, rather than a discrete action or a bounding box around a referred object.

2.2.2 Vision Encoders in Autonomous Navigation

Convolutional Neural Networks (CNNs) are widely used for visual recognition [41], including ResNets [42], DenseNets [43], EfficientNets [44], and VGG-style backbones [45]. In VLN, such encoders are often paired with language models such as Long Short-Term Memory (LSTM) networks [46] to follow instructions [3]. Vision Transformers (ViTs) treat an image as a sequence of patches and apply self-attention for more global context [47, 48].

Krantz et al. [4], in *Beyond the Nav-Graph: Vision-and-Language Navigation in Continuous Environments*, use a ResNet-50 visual encoder and argue that discrete navigation-graph assumptions can inflate reported performance. Controllable Imitative Reinforcement Learning (CIRL) similarly relies on vision encoders to learn driving policies from visual input [49]. Imaging laser radars have also been studied as complementary sensors for robotics perception [50].

For this thesis, the key takeaway is that strong visual features—and especially features that can be aligned with language—are a prerequisite for grounding navigable regions. Chapter 4 therefore uses CLIP visual features rather than a purely vision-only CNN backbone.

2.3 Basics of Natural Language Processing

Natural Language Processing (NLP) studies how computers interpret and generate human language [51, 52]. It combines computational linguistics with statistical and neural models.

2.3.1 Role of NLP in Autonomous Navigation

In language-guided navigation, NLP is required to interpret passenger or operator commands and relate them to the visual scene. Classic work by Chen and Mooney [53] showed how natural language instructions can be grounded into navigational behaviour. More recent VLN systems treat instruction following as a core multimodal problem rather than a pure control problem.

2.3.2 Basic Methods in NLP

Language models assign probabilities to word sequences and underpin many NLP systems [54, 55]. **Text classification** assigns documents or utterances to categories [55]. **Attention mechanisms** compute importance-weighted combinations of token representations and are central to modern sequence models and transformers [56, 48].

For outdoor driving commands, the linguistic input is typically short (a few words to a short sentence) but referentially rich: it names objects, colours, spatial relations, and manoeuvres. The system must therefore encode word-level semantics and align them with visual features, which is exactly the role of the CLIP language encoder used in Chapter 4.

2.3.3 Text Encoders in Autonomous Navigation

Text encoders such as word2vec, GloVe, and Bidirectional Encoder Representations from Transformers (BERT) convert language into vectors usable by downstream models [57, 55]. In VLN, Raychaudhuri et al. [58] propose language-aligned supervision for off-path recovery and a metric that counts completed sub-instructions. Dual-scale graph transformers such as DUET [59] combine coarse map reasoning with fine-grained language grounding for indoor VLN.

In our setting, we prefer a jointly trained vision–language encoder (CLIP) over a language-only encoder, because the command and the road scene must be compared in a shared embedding space before dense prediction.

2.4 Combining Vision and Language: Visual-Language Embeddings

Visual-language embeddings place images and text in a common representation space so that semantic matches lie close together [60, 61].

2.4.1 Introduction to CLIP and Beyond

CLIP (Contrastive Language-Image Pretraining) [2] is trained on large-scale internet image-text pairs. An image encoder and a text encoder are jointly optimised so that matched pairs are close in a shared embedding space (Figure 2.2). This dual-encoder design makes CLIP a natural backbone for VLN, where camera frames and linguistic commands must be aligned.

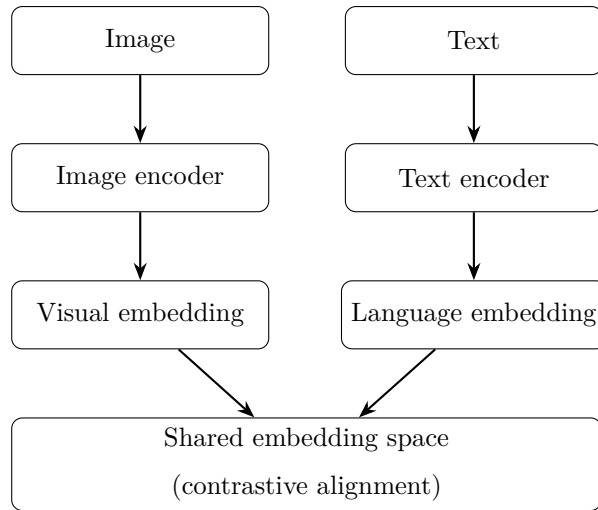


Figure 2.2: CLIP [2] encodes images and text into a shared embedding space via contrastive learning. We use CLIP features for both modalities in Chapter 4.

Larger generative models such as GPT-4 [62] and multi-modal architectures such as XDecoder [63] further expand the space of vision-language systems. They are not used in this thesis, but they illustrate the broader trajectory of the field toward richer multimodal reasoning.

2.4.2 Why They Matter for Language-Guided Navigation

In language-guided navigation, visual-language embeddings let a system relate what it sees to what it is told to do [5]. Dense-prediction adaptations of CLIP, such as converting image-text matching into pixel-text matching [64], are particularly relevant because our outputs are segmentation masks rather than classification scores.

The practical implication for Chapters 3 and 4 is architectural: we treat VLN as grounding a command into a navigable region using CLIP features, then hand that region to a planner, rather than translating language directly into a discrete action.

2.5 Language-Guided Navigation

2.5.1 Overview

Language-guided navigation develops systems that interpret natural language instructions and use them to move through a physical or virtual environment [3]. It combines NLP, computer vision, and decision-making. Challenges include:

1. **Ambiguity:** the same words can mean different regions depending on context.
2. **Variability:** different phrasings can describe the same manoeuvre.
3. **Multimodal integration:** language, vision, and control must be combined reliably.
4. **Real-time constraints:** decisions must keep up with a moving vehicle [3].

Applications range from passenger voice commands in autonomous vehicles, to service robots, assistive technologies, and interactive virtual environments [3]. For outdoor driving specifically, the hard case—and the focus of this thesis—is grounding intermediate navigable regions when the final referred goal is not yet visible.

2.5.2 Existing Datasets

Several datasets have shaped language-guided navigation research:

1. **Room-to-Room (R2R):** Anderson et al. [3] provide natural language instructions over Matterport3D indoor panoramas.
2. **Touchdown:** Chen et al. [5] target outdoor urban navigation with Google Street View and discrete actions.
3. **StreetNav:** Mirowski et al. [65] study instruction following on Google Street View environments.
4. **Mapillary Street-Level Sequences (MSLS):** Warburg et al. [66] provide large-scale street-level sequences for lifelong place recognition. Mapillary Vistas [67] is a related semantic segmentation dataset and should not be confused with MSLS.
5. **Active Vision Dataset (AVD):** Ammirato et al. [68] support active vision in indoor scenes.

6. **Stanford 2D-3D-Semantics (2D-3D-S)**: Armeni et al. [69] provide RGB-D indoor data with semantic annotations.
7. **AI2-THOR**: Kolve et al. [12] offer interactive photorealistic indoor rooms.
8. **House3D**: Wu et al. [70] provide large-scale diverse indoor 3D scenes for navigation and instruction following.

Table 2.1 positions the datasets most relevant to our contribution—including Talk2Car [15], Referring Navigable Regions (RNR) [1], and CARLA-NAV—along the axes that matter for outdoor language-guided driving.

Dataset	Environment	Dynamics	Action space	Annotation	Live eval.
R2R [3]	Indoor	Static scenes	Graph nodes	Path + instr.	No
Touchdown [5]	Outdoor	Static panoramas	Discrete actions	Path + instr.	No
StreetNav [65]	Outdoor	Static panoramas	Discrete actions	Path + instr.	No
Talk2Car [15]	Outdoor	Dynamic (nuScenes)	Object grounding	Object box	No
RNR [1]	Outdoor	Static frames	Region grounding	Navigable mask	No
CARLA-NAV (ours)	Outdoor	Dynamic traffic	Continuous (planner)	Navigable mask	Yes

Table 2.1: Comparison of language-guided navigation and grounding datasets along axes relevant to outdoor autonomous driving. CARLA-NAV uniquely combines outdoor dynamic scenes, continuous control via a local planner, navigable-region masks, and live episode-level evaluation. Details of CARLA-NAV are given in Chapter 3.

Most prior datasets either assume indoor graphs, restrict outdoor agents to discrete actions on pre-recorded panoramas, or ground objects rather than navigable road regions. Few support live episode completion in a fully navigable dynamic environment. These gaps motivate CARLA-NAV.

2.5.3 Existing Methods

Language-guided navigation methods are often grouped as modular, end-to-end, or hybrid [3, 5].

Modular pipelines separate language understanding, perception, and decision-making [71, 72, 73]. Specialisation helps, but errors can propagate across stages [74, 71].

End-to-end models map instructions and images directly to actions [75, 74, 76]. They can perform well with enough data, but are harder to interpret and debug.

Hybrid approaches combine specialised modules with learned policies [74, 75]. Training may use reinforcement learning, imitation learning, or both [3, 74].

Specific VLN methods include:

- **Sequence-to-sequence VLN:** Anderson et al. [3] map image sequences and instructions to actions with reinforcement learning from demonstrations and environment rewards.
- **Cross-modal matching:** Wang et al. [77] propose Reinforced Cross-Modal Matching (RCM) to align sub-instructions with observed scenes.
- **Progress monitoring and regret:** Ma et al. [78, 79] introduce a progress monitor, regret/backtracking, and visual-textual co-grounding so the agent can track instruction progress and decide whether to continue or reverse.
- **Environment Dropout (EnvDrop):** Tan et al. [80] randomly drop visual observations during training to improve robustness to environment change.
- **Weakly supervised mapping:** Chen et al. [81] build multi-granularity maps for object localisation and goal prediction on VLN-CE.

Despite this progress, outdoor dynamic driving with continuous control and interpretable region grounding remains comparatively underexplored. That is the setting addressed in Chapters 3 and 4.

2.6 Introduction to CARLA Simulator

CARLA (Car Learning to Act) is an open-source simulator for autonomous driving research [7]. It provides urban layouts, vehicles, pedestrians, configurable weather and sensors, and an API for controlling the simulation. A server-multi-client architecture supports scalable experiments. Sensors include cameras, LiDAR, depth, and GPS. Maps follow the OpenDRIVE standard; ScenarioRunner defines traffic situations; a ROS bridge is available. CARLA also ships baseline agents and supports record/replay for controlled comparisons.

These properties make CARLA suitable for CARLA-NAV: we can spawn dynamic traffic, collect synchronized camera and pose data, project clicked navigable regions into masks, and later evaluate live navigation rather than only scoring pre-recorded frames.

2.7 Summary

This chapter established the background needed for the remainder of the thesis. Autonomous driving combines perception, planning, and control; our contribution sits at the language-conditioned perception layer. Computer vision supplies dense prediction tools; NLP and visual-language embeddings—especially CLIP—supply the alignment between commands and images. Existing VLN datasets and methods largely assume indoor graphs, discrete outdoor actions, or static grounding, as summarised in Table 2.1. CARLA provides a controllable outdoor

simulator in which those assumptions can be relaxed. Chapter 3 introduces the CARLA-NAV dataset built on this foundation, and Chapter 4 presents the multi-task grounding network and live-navigation results.

Chapter 3

CARLA-NAV Dataset

The process of data collection is a critical component in the development of any machine learning model. The quality and relevance of the data collected directly influence the performance and effectiveness of the final model. In the context of autonomous driving, the data collection process becomes even more crucial due to the complexity and variability of real-world driving scenarios.

In this chapter, we describe the creation of CARLA-NAV, the meta-dataset that supports the language-guided navigation approach developed in this thesis.¹ We began with Grand Theft Auto V (GTA V) as an initial data collection environment, then switched to the research-oriented CARLA simulator [7]. We also explain our shift from predicting steering, acceleration, and brake values to predicting segmentation maps for navigation, detail the observer–navigator–verifier annotation protocol, and summarize dataset statistics and challenges.

3.1 Introduction

The ability to navigate based on linguistic descriptions of the environment is a remarkable human capability, a direct consequence of associating visual elements with their linguistic descriptions. Referring Expression Comprehension (REC) and Referring Image Segmentation (RIS) aim to localize visual objects based on linguistic descriptions [9, 8]. However, these localizations are not directly usable for navigation, as they do not answer “which region” on the road to navigate to. Referring Navigable Regions (RNR) [1] was proposed to localize navigable regions in a static front-camera image corresponding to a linguistic command.

While single image-based visual grounding methods showcase the ability of neural models to correlate visual and linguistic data, they are limited in several ways. They are trained on carefully paired data, assuming that the region to be grounded is always visible in the frame. They give erroneous output when the region is not currently visible, is occluded, or goes out of

¹Parts of this chapter appear in Jain et al., *Ground then Navigate: Language-guided Navigation in Dynamic Scenes*, IEEE ICRA 2023 [82].

frame. Such scenarios are part of everyday language-guided navigation; for instance, in “Take a right once you see the traffic signal,” the traffic signal may not be immediately visible.

Another major limitation is that single-image predictions are devoid of temporal context, which is crucial in successful navigation in dynamically changing environments. Finally, since single-image methods are often evaluated on frames from pre-recorded videos, they cannot be validated on their ability to complete an entire episode from start to finish.

Most existing works on language-guided navigation make assumptions that limit applicability in real-world outdoor settings. Indoor methods often assume a fully known environment that can be discretized into a graph [3, 12]. Outdoor methods such as Touchdown [5] allow navigation by choosing from a small discrete action set on pre-recorded street views. This discretization limits fine manoeuvres such as “park between the two cars on the right.”

To address these limitations, we created CARLA-NAV using the CARLA simulator [7]. Combined with a visual grounding-based approach, the dataset allows fine-grained control over the vehicle by enabling navigation to any drivable region on the road. To the best of our knowledge, this is among the first attempts toward live language-based navigation in outdoor environments.

3.2 Initial Data Collection with GTA V

We initially attempted to collect data using Grand Theft Auto V (GTA V) and the DeepGTAV and V-Pilot mods. The game’s realistic graphics, diverse urban environments, and dynamic traffic made it an appealing choice for simulating real-world driving conditions. DeepGTAV allows extraction of camera images, vehicle state, and information about other agents in the scene.

Our initial approach was to train a deep learning model to predict steering, acceleration, and brake. However, we encountered several challenges. First, predicting only three continuous controls without sufficient high-level context proved brittle, consistent with observations that end-to-end driving models can struggle without structured guidance [83, 74]. Second, supervised learning alone was insufficient without a carefully designed reward if we moved to reinforcement learning, and defining rewards for language-conditioned driving is itself a hard problem [84, 85]. GTA V also did not provide the per-pixel and camera-calibration access needed for dense navigable-region annotation.

Given these challenges, we switched to predicting segmentation masks under supervised learning. Dense mask prediction provides a more interpretable target and a clearer visual representation of what the model believes is navigable [86, 87]. In the next section, we discuss why CARLA made this approach feasible.

3.3 Switching to CARLA for Data Collection

After encountering limitations with GTA V, we switched to CARLA [7], an open-source simulator designed for autonomous driving research. CARLA offers several advantages that made it suitable for our purposes:

1. **Research-oriented control:** CARLA provides flexible control over weather, time of day, and the behaviour of other vehicles and pedestrians, which is crucial for a diverse dataset.
2. **Consistent physics:** Vehicle motion and interactions with the environment are more reliable than in a commercial game pipeline.
3. **Rich environment state:** Access to 3D object positions, road layout, and sensor meta-data supports precise annotation.
4. **Camera calibration:** Live camera intrinsics and orientation matrices enable transforming 2D screen coordinates into 3D world coordinates—essential for projecting navigable regions and driving a local planner.

These capabilities allowed us to adopt visual grounding: associating linguistic descriptions of navigable regions with their visual counterparts and predicting a segmentation map that can be used for navigation. This shift would not have been feasible in GTA V due to limited access to internal game data.

3.4 Data Collection Methodology

The data collection process in CARLA was a collaborative effort involving three roles: the observer, the navigator, and the verifier.

- **Observer:** Provides clear linguistic commands based on the current environment state. Commands are generated on-the-fly rather than scripted, capturing natural variation.
- **Navigator:** Drives in CARLA by clicking navigable regions in the front-camera view and follows the observer’s command as naturally as possible. A restart option is available for erroneous clicks.
- **Verifier:** Reviews the language command, navigated path, and generated masks, and decides whether the episode is included in the dataset.

The 2D screen coordinates clicked by the navigator are transformed into 3D world coordinates using the inverse projective transform with CARLA’s camera intrinsic and orientation

matrices. This 3D position is passed to the local planner (CARLA’s default planner in our case; the design is modular and can later use planners such as NEAT [88]).

An episode comprises multiple mouse clicks until the final navigable region becomes visible in the front view. Intermediate clicks signify intermediate navigable regions; the last click depicts the final goal region. Mouse clicks are converted into segmentation masks by drawing a $3\text{m} \times 4\text{m}$ rectangle (approximately the size of a car) in the top view, centred at the click, and projecting that rectangle into the front camera. Overall, only language commands and mouse clicks require manual effort; the rest of the annotation process is automated.

For the trajectory prediction task used later for interpretability, we take the 3D position of the vehicle in successive frames and project those positions onto the front-camera image. Trajectory prediction is treated as a dense prediction task and is not quantitatively evaluated.

Figure 3.1 shows ground-truth annotations for three sampled frames from an episode of CARLA-NAV.

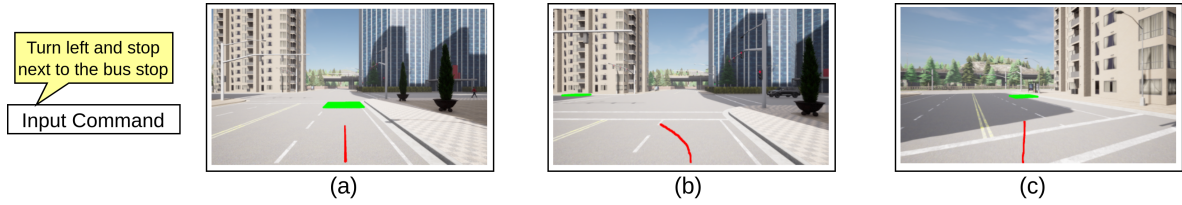


Figure 3.1: Ground truth annotations of three sampled frames from an episode of the CARLA-NAV dataset. The textual command for the episode is shown on the top. The green rectangle illustrates the navigable region, and the red curve corresponds to the short future trajectory. (a) At the start, the left turn or the final navigable region is not visible, so a straight path is chosen as the intermediate mask; (b) the intermediate mask corresponding to the left turn; (c) the final navigable region (stop next to the bus stop).

3.5 Dataset Characteristics

Relative to prior language-guided navigation and grounding datasets surveyed in Chapter 2 (Table 2.1), CARLA-NAV is designed for outdoor dynamic scenes with continuous control and live episode-level evaluation. CARLA-NAV contains episodic data: each episode consists of a language command and the corresponding CARLA video of navigation toward the final goal region described by the command. The ground-truth segmentation mask for each frame corresponds to either the final or an intermediate navigable region. Each frame is additionally annotated with a plausible future trajectory of the vehicle in the next few frames.

The dataset includes video sequences captured in 8 different maps, 14 distinct weather conditions, and a diverse range of dynamically moving vehicles and pedestrians. Language commands contain detailed visual descriptions and describe a wide range of manoeuvres. Commands can have a single manoeuvre (e.g., park behind the black car on the left) or multiple manoeuvres (e.g., take a right turn and park near the bus stand), with a maximum of three manoeuvres.

Table 3.1 reports dataset statistics. Since each episode contains multiple frames, the overall dataset size is 83,297 frames across all splits. During training we use pre-recorded sequences. During inference on validation and test splits, the vehicle is spawned at the corresponding starting location and navigation is performed live based on network prediction, not on the pre-recorded sequences.

Split	# Episodes	# Frames	Command length (words)	Clicks
Train	500	75,010	6.92	1.94
Val	25	3,300	6.76	2.00
Test	34	4,987	7.44	1.97

Table 3.1: Dataset statistics for the CARLA-NAV dataset. Except “# Episodes” and “# Frames”, average values per episode are reported for the other columns.

Commands mainly consist of up to three subcommands, most having two. Subcommands include turning, going straight, stopping, and parking, and reference pedestrians, infrastructure, road elements, other vehicles, and scenery.

3.6 Data Collection Challenges

We encountered several practical challenges during data collection:

1. **Asynchronous simulation:** CARLA operates asynchronously, so the simulator does not wait for annotation code to finish before advancing. Careful synchronization was required so that frames being annotated matched the displayed state.
2. **Slopes:** Transforming 2D screen coordinates into 3D world coordinates assumes a flat ground plane. Transitions from slopes to flat surfaces introduced inaccuracies that required special handling.
3. **Split contamination:** Given the finite CARLA maps, similar scenarios can appear across train, validation, and test splits. We mitigated this by shuffling episodes, randomizing spawn locations, and verifying diversity across splits.

4. **Subjectivity:** Different observers and navigators may interpret the same command differently. Role rotation and verifier checks reduced individual bias.

3.7 Data Collection Team and Conclusion

The CARLA-NAV dataset was curated collaboratively. The team comprised KV Aditya Srivatsa, Amogh Tiwari, and myself, with each of us rotating through the roles of observer, navigator, and verifier. Dr. Vineet Gandhi also participated, providing more complex linguistic commands and acting as a verifier in ambiguous cases.

In conclusion, CARLA-NAV provides a diverse set of language-guided outdoor navigation episodes with dense navigable-region and trajectory annotations. The dataset and collection toolkit form the foundation for the multi-task network and live-navigation experiments presented in Chapter 4.

Chapter 4

Vision-and-Language Navigation in Autonomous Driving

This chapter presents our approach to Vision-and-Language Navigation (VLN) for autonomous driving in dynamic outdoor environments.¹ Building on the CARLA-NAV dataset introduced in Chapter 3, we formalize the problem, describe a multi-task grounding network, explain live navigation, and report quantitative and qualitative results. Dataset creation details are not repeated here; we focus on modelling, inference, and evaluation.

4.1 Background and Related Work

Visual grounding associates linguistic descriptions with visual counterparts via proposal-based Referring Expression Comprehension (REC) [9, 11, 89, 90, 91, 92, 93, 94] or pixel-level Referring Image Segmentation (RIS) [95, 96, 97, 98]. Referring Navigable Regions (RNR) [1] localizes navigable road regions for language commands, but is limited to static images in pre-recorded sequences [99].

Indoor VLN has benefited from interactive environments [12, 70, 13] and methods based on imitation learning [100, 101], behaviour cloning [102], sequence-to-sequence translation [3], and cross-modal attention [103]. Many approaches assume known topologies and discrete navigation graphs; Krantz et al. [4] argue that nav-graph assumptions can inflate performance, and Hu et al. [104] question the role of vision when route structure alone can succeed in unseen environments.

Outdoor VLN on Google Street View and Touchdown [5, 105, 106] typically uses discrete actions [6, 10, 107]. CARLA-based language navigation has been limited in action diversity [14], and Talk2Car focuses on object localization [15]. We instead reformulate RNR for dynamic outdoor settings and perform live navigation with continuous control via a local planner.

¹This chapter is based on Jain et al., *Ground then Navigate: Language-guided Navigation in Dynamic Scenes*, IEEE ICRA 2023 [82].

4.2 Problem Statement

Given video frames $V = \{v_{t-k}, v_{t-k+1}, \dots, v_t\}$, a contextual historical trajectory P , and a language command $L = \{l_1, l_2, \dots, l_N\}$, where t is the current timestamp, k is the historical window size, and N is the maximum number of words, the goal is to predict the navigable mask y_t and the future trajectory mask z_t for the current frame v_t .

The contextual trajectory P provides information about which part of a multi-step command has already been executed. For example, for “turn left and park near the blue dustbin,” P indicates whether the left turn has already been taken. The spatial location of the navigation mask should determine the trajectory path’s direction, and vice versa.

4.3 Methodology

We propose a multi-task network for navigation-region prediction and future-trajectory prediction. Both tasks are treated as dense prediction tasks for interpretability. During live inference, dense pixel points are converted to 3D world coordinates using inverse projective transformation. The architecture is illustrated in Figure 4.1.

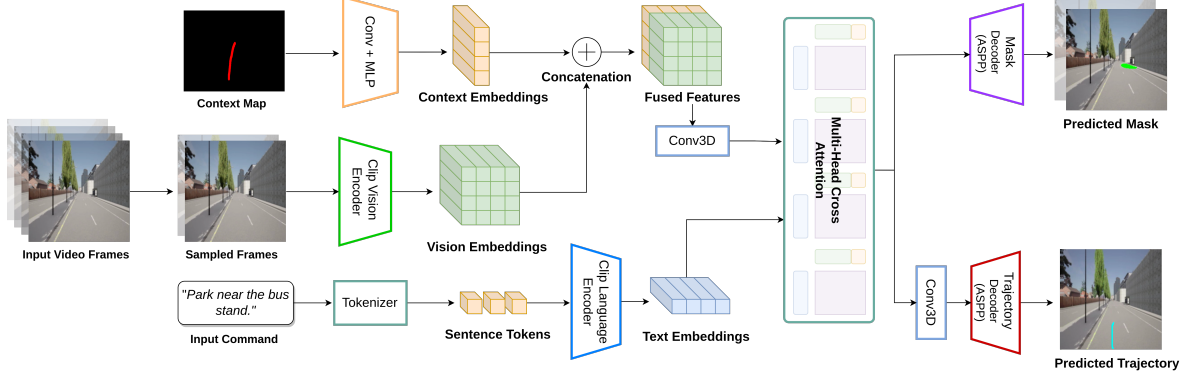


Figure 4.1: Overall pipeline of the proposed approach. Given the visual frames, already executed past trajectory (context map), and the textual command, the network predicts a segmentation map corresponding to the navigable region and a plausible future trajectory.

4.3.1 Feature Extraction

We utilize CLIP [2] to extract linguistic and visual features. For the linguistic expression L , we tokenize the command with the CLIP tokenizer and obtain word-level features $F^l = \{f_1^l, \dots, f_N^l\}$ of shape $\mathbb{R}^{N \times C_l}$. For visual frames V , CLIP encodes video features $F^v = \{f_{t-k}^v, \dots, f_t^v\}$. For trajectory context P , we project the past trajectory onto an image of the

same size as v_t and pass it through convolution and MLP layers to obtain F^p with the same feature size as the per-frame visual features.

4.3.2 Multi-Modal Fusion

The trajectory context feature $F^p \in \mathbb{R}^{C_v \times 1 \times HW}$ is concatenated with video features F^v along the temporal dimension, yielding $F^{vp} \in \mathbb{R}^{C_v \times (T+1) \times HW}$. We then apply multi-head self-attention over the joint contextual feature F^{vp} and linguistic feature F^l :

$$\begin{aligned} F &= F^{vp} \odot F^l \\ A &= \text{Mhead}(F, F, F) \\ M &= \text{Conv3D}(A * F) \end{aligned} \tag{4.1}$$

where $\odot$ denotes length-wise concatenation of word-level linguistic features with the joint visual-trajectory feature, Mhead is multi-head self-attention, $*$ denotes matrix multiplication, and Conv3D collapses the temporal dimension. M is the final multi-modal contextual feature.

4.3.3 Segmentation Heads and Loss

Separate segmentation heads (sequences of convolutions with upsampling) predict the navigation and trajectory masks from M . We train with combo loss [108], a combination of binary cross-entropy and Dice loss:

$$\begin{aligned} L_{\text{bce}} &= -(y_t \log(\hat{y}_t) + (1 - y_t) \log(1 - \hat{y}_t)) \\ L_{\text{dice}} &= 2 \cdot \frac{\hat{y}_t \cap y_t}{\Sigma \hat{y}_t + \Sigma y_t} \\ L_{\text{combo}} &= \lambda L_{\text{bce}} - (1 - \lambda) L_{\text{dice}} \end{aligned} \tag{4.2}$$

with $\lambda = 0.3$. The approach is end-to-end trainable; the predicted trajectory correlates with the predicted navigation mask and indicates the vehicle’s short-term route.

4.3.4 Live Navigation

To use the predicted navigable-region mask for navigation, we take the largest connected component and use its centroid as the target point for the local planner. As the vehicle approaches the final region, the predicted mask area tends to increase due to perspective. If the mask area exceeds an empirically chosen threshold for five consecutive predictions, we treat the region as the final goal and stop. During inference we predict every 5 frames; training uses every frame.

4.4 Evaluation Metrics

Following prior VLN work [6, 107, 5], we use **Task Completion (TC)** as the primary success metric: whether the agent successfully navigates to the goal location for a given language command. We additionally use **Fréchet Distance** [109, 110] and **normalized Dynamic Time Warping (nDTW)** [111] to compare the live predicted path with the ground-truth path.

Fréchet Distance and nDTW provide useful path-similarity signals, but they are not fully robust indicators of average navigation quality: a single early wrong turn can dominate the score, and a linguistic command may admit multiple valid trajectories (e.g., “park near the blue dustbin”). Task Completion therefore remains the primary metric.

4.5 Experimental Results

4.5.1 Implementation Details

We use a CLIP backbone [2] for feature extraction. Frames are selected with stride 10 and resized to 224×224 . Per-frame visual features have spatial resolution $H = W = 7$ and channel dimension $C_v = 512$. The historical contextual trajectory is plotted from episode start to the current timestamp, resized to 640×480 , and encoded to match the visual feature resolution. Linguistic features use the CLIP tokenizer and language encoder with maximum command length $N = 20$ and $C_l = 512$. We train with batch size 32 and AdamW [112], initial learning rate 10^{-4} , and polynomial learning-rate decay with power 0.5. For combo loss, $\lambda = 0.3$. All methods and baselines were trained from scratch on CARLA-NAV.

4.5.2 Baselines and Ablations

We compare our proposed approach CLIP-MC (multiple frames with contextual trajectory) against:

- RNR-S / RNR-SC: the RNR approach of Rufus et al. [1] with and without contextual trajectory, trained from scratch on CARLA-NAV;
- CLIP-S / CLIP-SC: single-frame CLIP variants without and with contextual trajectory;
- CLIP-M: multiple frames without contextual trajectory;
- CLIP-MC: multiple frames with contextual trajectory (proposed).

Table 4.1 reports Task Completion. CLIP-MC outperforms all variants. Introducing historical contextual trajectory improves validation TC by 4 percentage points for single-frame

approaches (CLIP-SC 0.52 vs. CLIP-S 0.48) and by 16 percentage points for multi-frame approaches (CLIP-MC 0.72 vs. CLIP-M 0.56). The multi-frame approach CLIP-M improves by 8 percentage points over CLIP-S on both validation and test splits (0.56 / 0.55 vs. 0.48 / 0.47). These results indicate that multiple frames and contextual trajectory are both required for effective outdoor VLN. CLIP-based single-frame models also outperform non-CLIP RNR variants on the test split, highlighting the benefit of CLIP multi-modal features.

Method	Task Completion	
	Val	Test
RNR-S	0.44	0.29
RNR-SC	0.52	0.32
CLIP-S	0.48	0.47
CLIP-SC	0.52	0.50
CLIP-M	0.56	0.55
CLIP-MC	0.72	0.68

Table 4.1: Results on the Task Completion metric. The superior performance of CLIP-MC showcases the effectiveness of historical context for the navigation task.

Table 4.2 reports Fréchet Distance (lower is better) and nDTW (higher is better). CLIP-MC again outperforms all other variants by clear margins.

Method	Fréchet Distance ↓		nDTW ↑	
	Val	Test	Val	Test
RNR-S	28.14	42.45	0.35	0.16
RNR-SC	21.64	44.65	0.45	0.33
CLIP-S	40.30	42.53	0.23	0.24
CLIP-SC	35.58	38.49	0.36	0.39
CLIP-M	32.92	53.10	0.39	0.26
CLIP-MC	13.54	15.06	0.54	0.59

Table 4.2: Experimental results on Fréchet Distance and nDTW. ↓ indicates lower is better and ↑ indicates higher is better.

Table 4.3 studies the effect of the number of video frames for multi-frame models on Task Completion. Performance generally improves as more frames are used; the best result is obtained with $n = 8$ frames for CLIP-MC on both validation and test.

Method	Split	Task Completion				
		$n = 1$	$n = 2$	$n = 4$	$n = 6$	$n = 8$
CLIP-MC	val	0.52	0.48	0.52	0.68	0.72
	test	0.50	0.53	0.56	0.62	0.68
CLIP-M	val	0.48	0.44	0.48	0.52	0.56
	test	0.47	0.47	0.50	0.53	0.55

Table 4.3: Ablation on the number of frames for multi-frame models on the Task Completion metric.

4.5.3 Qualitative Results

Figure 4.2 compares CLIP-MC with RNR-S during live inference. Paths are overlaid on the aerial map of the CARLA environment. With multi-frame and historical trajectory context, CLIP-MC successfully performs turning and stopping manoeuvres where the static-image RNR baseline fails. For “change to the left lane,” RNR-S continues straight while CLIP-MC changes lane (with a slight delay). In a curved-road case near a traffic light, both methods stop early, mistaking the curve for an intersection—an instructive failure mode.

4.6 Failure Modes

Qualitative inspection of live episodes, including the cases shown in Figure 4.2, reveals recurring failure patterns:

1. **Delayed lane change:** On commands such as “change to the left lane,” CLIP-MC eventually executes the manoeuvre but with a slight delay relative to the ground-truth path, while RNR-S may continue straight. Temporal context helps, but reaction lag remains.
2. **Curve mistaken for an intersection:** On a curved road approaching a traffic light, both CLIP-MC and RNR-S stop prematurely, treating the geometry as an intersection. This suggests sensitivity to road layout cues that resemble junction structure.

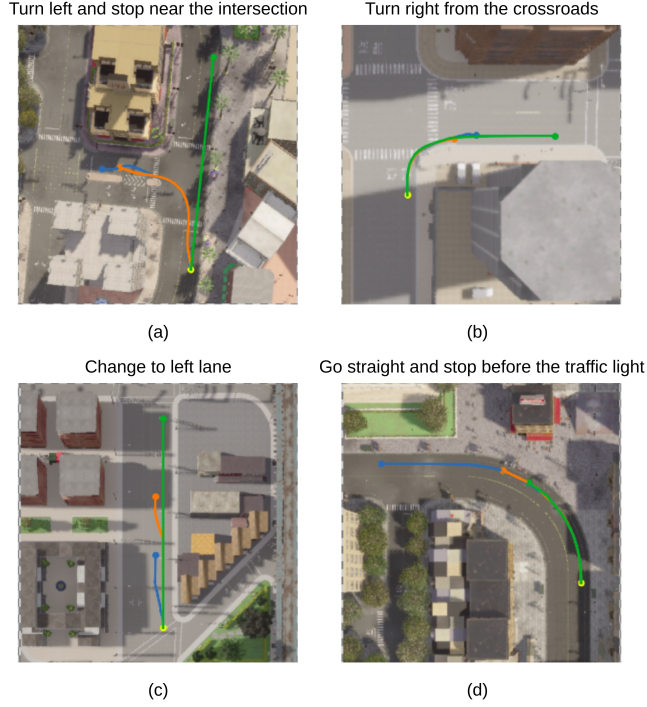


Figure 4.2: Qualitative navigation results on CARLA-NAV. Yellow marks the start. Orange depicts the path taken by CLIP-MC, green denotes RNR-S, and blue represents the ground-truth path.

3. **Early stopping from the area threshold:** When several candidate goal regions are visible, the area-based stopping rule can fire before the intended final region is reached, because a large connected component satisfies the threshold for five consecutive predictions.
4. **Referential ambiguity and path metrics:** Commands such as “park near the blue dustbin” admit multiple valid stopping locations. Fréchet Distance and nDTW then penalise a spatially plausible path that differs from the recorded demonstration, even when Task Completion may still succeed.
5. **Synthetic-to-real domain gap:** All training and evaluation are in CARLA. Appearance, sensor noise, and traffic behaviour in the real world differ, so transfer without domain adaptation [113, 114] is an open risk.

4.7 Discussion

Historical contextual trajectory informs the model which sub-instructions have already been executed, yielding large gains for multi-frame models. Multiple frames capture broader temporal visual context needed in dynamic scenes. CLIP provides stronger multi-modal features than a ResNet-style RNR backbone for aligning language with road imagery.

The area-based stopping criterion is simple and effective in many cases, but can be suboptimal when multiple candidate goal regions exist. Future improvements include confidence-aware stopping, proximity to the goal, temporal consistency of the mask, and semantic confirmation of referred objects [107]. Domain adaptation toward real-world imagery [113, 114] is another important direction.

4.7.1 Limitations of the Current Evaluation

Several analyses that would strengthen the empirical story are not reported in this thesis, because they were not part of the original ICRA evaluation protocol:

- **No per-manoeuvre breakdown:** CARLA-NAV commands contain up to three sub-commands spanning turning, going straight, stopping, and parking, yet Task Completion and path metrics are reported only in aggregate.
- **Unevaluated trajectory head:** Future-trajectory prediction is used for interpretability and is never scored quantitatively against ground truth.
- **No runtime profiling:** Live navigation is demonstrated in CARLA, but wall-clock inference latency and frame rate are not reported, so the real-time claim is not numerically substantiated.

Stating these gaps explicitly is preferable to implying that the current metrics fully characterise system behaviour. They are natural targets for follow-up experiments.

4.8 Chapter Summary

We reformulated RNR for dynamic outdoor VLN, introduced a multi-task grounding network with CLIP features, multi-frame context, and historical trajectory, and demonstrated live navigation in CARLA. Quantitative results show that CLIP-MC substantially outperforms single-frame and non-contextual baselines on Task Completion, Fréchet Distance, and nDTW. Failure modes and evaluation limitations clarify where the approach still falls short. The next chapter concludes the thesis and outlines future work.

Chapter 5

Conclusion

This thesis addressed Vision-and-Language Navigation for autonomous driving in dynamic outdoor environments. We argued that existing approaches—graph-based indoor VLN, discrete-action outdoor VLN, and single-image navigable-region grounding—are insufficient when the referred goal is not yet visible, when fine manoeuvres are required, and when navigation must be evaluated live in a changing scene.

We introduced CARLA-NAV, a meta-dataset of language-guided outdoor navigation episodes in the CARLA simulator, with dense annotations for intermediate and final navigable regions and short-term future trajectories. The observer–navigator–verifier protocol, combined with automated projection of clicked regions into segmentation masks, enabled scalable curation across multiple maps and weather conditions.

We then proposed a multi-task network that predicts navigable-region and future-trajectory masks from multi-frame CLIP visual features, a historical contextual trajectory, and the language command. Predicted masks are used directly for live navigation via a local planner and an area-based stopping criterion. Experiments showed that combining multiple frames with historical trajectory context yields substantial gains: CLIP-MC reaches Task Completion of 0.72 / 0.68 on validation / test, outperforming single-frame CLIP and RNR baselines, with corresponding improvements in Fréchet Distance and nDTW.

Overall, grounding-then-navigating provides an interpretable and continuous-control alternative to discrete VLN formulations, and is a promising direction for language-conditioned autonomous driving.

5.1 Future Work

- **Real-world transfer:** CARLA-NAV is synthetic. Domain adaptation techniques [113, 114] are needed before deployment on real vehicles.
- **Learnable stopping:** Replacing the rule-based area threshold with a learned stopping policy [107] should improve robustness.

- **Trajectory-conditioned control:** The predicted future trajectory is currently used for interpretability; incorporating it into the planner could improve end-to-end behaviour.
- **Richer evaluation:** Developing metrics that respect multiple valid paths for the same command would better reflect outdoor driving ambiguity.
- **Interactive language:** Extending the setting to clarification dialogues and multi-turn instructions would bring the system closer to human co-driving.

Related Publications

1. Jain, K., Chhangani, V., Tiwari, A., Krishna, K. M., & Gandhi, V. (2023). Ground then Navigate: Language-guided Navigation in Dynamic Scenes. *IEEE International Conference on Robotics and Automation (ICRA) 2023*.

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